A "fact" about fractions

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I was checking my work on a problem recently[1], where I reached the equation

$$-\frac{1}{8}-\frac{3}{4}-\frac{3}{2}+\frac{1}{24}\stackrel{?}{=}\frac{7}{3}.$$

At this point I concluded that I must have made a mistake. Without even calculating, I knew the left side had denominator 24, but the right side had denominator 3.

Or so I thought. Actually, the two sides are equal. Is it intuitive that the denominator, in lowest terms, can decrease? Maybe. The example

$$\frac{1}{3}+\frac{1}{6}=\frac{1}{2}$$

is fairly common.

But I had to ask: why did I make this wrong assumption? Is there some other true statement we can conclude?

An Investigation

Question.

Are there integers a, b, c that make the following equation true?

$$8a+4b+2c+1=56$$

Clearly, the answer is no: taking both sides modulo 2 gives $0 \equiv 1 \pmod 2$, which is always false. This idea, wrongly applied to the fractional example, caused me to dismiss a correct solution.

But, returning to the original problem, there is something we can say:

Valid Denominators.

Find all possible denominators, in simplest form, of

$$\frac{a}{8}+\frac{b}{4}+\frac{c}{2}+\frac{d}{24}$$

where $a, b, c, d \in \mathbb{Z}$.

The answer is **all divisors of 24.** Notice that

$$\frac{a}{8}+\frac{b}{4}+\frac{c}{2}+\frac{d}{24}=\frac{3a+6b+12c+d}{24},$$

and the numerator can be any integer. So, the possible denominators are those that divide 24.

In general, given an expression $\frac{p_1}{q_1}+\frac{p_2}{q_2}+\cdots+\frac{p_n}{q_n}$, all we can say is that the denominator in lowest terms is a divisor of $\operatorname{lcm}(q_1, \ldots, q_n)$.

This isn't a very strong statement, but at least it's something. For example, we can quickly show that the following equation has no solutions for $a, b \in \mathbb{Z}$.

$$\frac{a}{8}+\frac{b}{3}=\frac{1}{48}$$

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1. The problem was finding the partial fraction decomposition of $\frac{1+x+x^2}{(1-x)^3(1+x)}$. I found the correct answer and plugged in $x=2$ to check, leading to the main equation. ↩